Polynomial, Determinant, and Formal-Derivative Identities for an Explicit Sextic Section

Lean Stage 1 Algebraic Certificates

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Abstract

We record seven kernel-checked commutative-algebra lemmas formalized in Lean 4 in the module `Hodge.ExplicitResidueAlgebra`. The lemmas concern an exact two-pair decomposition, a binary split factorization, two 4×4 Jacobian determinant factorizations, nonvanishing of every root of an explicit degree-24 section modulus in characteristic zero, nonvanishing of one leading Wronskian coefficient at every such root, and an identity expressing the binary Wronskian of two explicit binary forms via formal multivariate polynomial derivatives. The statements are mapped to Lean declarations below. They do *not* construct Betti or Hodge cohomology, cycle classes, geometric ideals, or a residue-cycle theorem. Toolchain pins, a theorem-to-declaration table, reproduction commands, and precise formalization boundaries are included. Companion Stage 2 nonvanishing certificates appear in a separate note.

Contents

1 Scope and non-claims **2**

2 Notation and standing hypotheses **2**

3 Main results **3**

4 Theorem–declaration mapping **4**

5 Axiom audit and verification evidence **5**

6 Toolchain pins and reproducibility **5**

7 Limitations and relation to the geometric program **6**

1 Scope and non-claims

Let $X \subset \mathbb{P}^5$ denote the complex projective hypersurface cut out by the homogeneous sextic

$$\sum_{j=0}^{2} (x_j^5\, x_{j+3} + x_{j+3}^6) = 0 \tag{1}$$

(indices understood as in the repository's explicit-curve encoding). The classical manuscript and Python residue certificate in the HODGE repository argue that certain complete intersection surfaces on X project nontrivially onto specified Griffiths Jacobian characters [1]. The present note formalizes only a bounded algebraic fragment of that route.

What this note proves. Exactly the seven Lean theorems listed in Section 4, as stated in `Hodge/ExplicitResidueAlgebra.lean` at mathematical source pin `e207394a9ed4255ad804860b8f69fde7e940dd54`.

What this note does not prove.

- The Hodge conjecture for X, in any rational or integral form, and not the general Hodge conjecture.
- Geometric cycle classes, Betti cohomology, Hodge structures, or a Lean realization of Villaflor Loyola's residue-cycle theorem [4].
- An actual Jacobian-quotient vector-space basis, reducedness or saturation of geometric ideals (Q, C) or (L, Q, C), or Fourier/Galois orbit spanning of rational Hodge classes.
- The Stage 2 simultaneous nonvanishing of all eight Wronskian/U coefficients and the discriminant $b^2 - 4c$ (treated in the companion note), and the tenth Python H/G-resultant certificate (still not Lean-proved).

These boundaries follow the repository records `docs/FULL_FORMALIZATION_STAGE1.md` and `docs/FULL_FORMALIZATION_STAGE2.md` [1].

2 Notation and standing hypotheses

Definition 1 (Section binary forms)**.** For a commutative ring R and coefficients $b, c, d, e, g \in R$, write

$$H(b, c; X, Y) = X^2 + b\,XY + c\,Y^2, \tag{2}$$

$$G(d, e, g; X, Y) = X^3 + d\,X^2Y + e\,XY^2 + g\,Y^3, \tag{3}$$

as in `Hodge.ExplicitCurve.H` and `Hodge.ExplicitCurve.G`.

Definition 2 (Section modulus)**.** Over any field K of characteristic zero, the scalar section modulus is the degree-24 monic polynomial identity

$$k^{24} + 66081312000\,k^{18} - 4811512860000\,k^{12} + 1167566400000\,k^{6} + 583200000 = 0. \tag{4}$$

The Lean definitions `ExplicitCurve.sb`, `sc`, `sd`, `se`, `sg` evaluate five explicit rational polynomials in k (recorded in `Hodge/ExplicitCurve.lean`). Stage 1 uses the already checked relation

$$2 \cdot \mathrm{sd}(k) - 3 \cdot \mathrm{sb}(k) - k = 0 \tag{5}$$

(`ExplicitCurve.rel_h5`) whenever (4) holds.

Remark 3. The docstring of `ExplicitCurve.m` describes m as irreducible over $\mathbb{Q}$. That irreducibility claim is classical/Python context and is *not* among the seven Lean theorems of this note. Stage 2 proves only $\mathrm{degree}(m) = 24$ in Lean.

3 Main results

Throughout, R is a commutative ring and K is a field of characteristic zero (with a $\mathbb{Q}$-algebra structure when section coefficients appear).

Theorem 4 (Two-pair decomposition)**.** *For $h_1, g_1, h_2, g_2, i \in R$ with $i^2 = -1$,*

$$\begin{aligned}(h_2 + h_1)(-(h_2^2 - h_2h_1 + h_1^2)) + (g_2 + ig_1)(g_2 - ig_1)\\ = (g_1^2 - h_1^3) + (g_2^2 - h_2^3).\end{aligned}$$

Lean proof outline. `linear_combination` against $-(g_1^2)\,(i^2 + 1)$. Declaration: `Hodge.ExplicitResidueAlgebra.two_pair_decomposition`. □

Theorem 5 (Split-pair factorization)**.** *For $u, v \in R$,*

$$(u + v) \cdot (v(u^4 - u^3v + u^2v^2 - uv^3 + v^4)) = u^5v + v^6. \tag{6}$$

Lean proof outline. Immediate ring normalization (`ring`). Declaration: `split_pair_decomposition`. □

Theorem 6 (Outer Jacobian determinant)**.** *For $h_1, h_2, i \in R$,*

$$\det \begin{pmatrix} 1 & 0 & 1 & 0 \\ h_2 - 2h_1 & 0 & h_1 - 2h_2 & 0 \\ 0 & i & 0 & 1 \\ 0 & -i & 0 & 1 \end{pmatrix} = 6i(h_2 - h_1). \tag{7}$$

Lean proof outline. Expand via `Matrix.det_succ_row_zero` and finish by `ring`. Declaration: `outer_jacobian_det`. No square-root hypothesis is used. □

Theorem 7 (Paired Jacobian determinant)**.** *For $h_1, h_2, i \in R$ and eight first partials*

$$h_{1x},\ h_{1y},\ g_{1x},\ g_{1y},\ h_{2x},\ h_{2y},\ g_{2x},\ g_{2y} \in R,$$

$$\det \begin{pmatrix} h_{1x} & h_{1y} & h_{2x} & h_{2y} \\ (h_2-2h_1)h_{1x} & (h_2-2h_1)h_{1y} & (h_1-2h_2)h_{2x} & (h_1-2h_2)h_{2y} \\ ig_{1x} & ig_{1y} & g_{2x} & g_{2y} \\ -ig_{1x} & -ig_{1y} & g_{2x} & g_{2y} \end{pmatrix}$$
$$= 6i(h_2-h_1)\,(h_{1x}g_{1y}-h_{1y}g_{1x})\,(h_{2x}g_{2y}-h_{2y}g_{2x}).$$

Lean proof outline. Same determinant expansion and ring normalization as Theorem 6. Declaration: `paired_jacobian_det`. Linking the eight symbols to actual polynomial derivatives is a separate obligation and is not discharged by this identity alone. □

Theorem 8 (Section roots are nonzero)**.** *If $k \in K$ satisfies* (4)*, then $k \neq 0$.*

Lean proof outline. Substitute $k = 0$ into (4) and contradict by `norm_num`. Declaration: `section_root_ne_zero`. □

Theorem 9 (Leading W coefficient)**.** *Assume moreover that K carries a $\mathbb{Q}$-algebra structure. If $k \in K$ satisfies* (4)*, then*

$$2 \cdot \mathrm{sd}(k) - 3 \cdot \mathrm{sb}(k) \neq 0. \tag{8}$$

Lean proof outline. From `ExplicitCurve.rel_h5`, $2 \cdot \mathrm{sd}(k) - 3 \cdot \mathrm{sb}(k) = k$; apply Theorem 8. Declaration: `leading_W_ne_zero`. This is one of eight coefficient nonvanishings required by the classical residue route; the remaining seven and the discriminant are Stage 2. □

Theorem 10 (Binary Wronskian via formal derivatives)**.** *For $b, c, d, e, g \in R$, let $x = X_0$, $y = X_1$ in $R[X_0, X_1]$, and set*

$$H_p = x^2 + C(b)\,xy + C(c)\,y^2,$$
$$G_p = x^3 + C(d)\,x^2y + C(e)\,xy^2 + C(g)\,y^3.$$

Then the formal partial Wronskian expands as

$$\partial_0 H_p \cdot \partial_1 G_p - \partial_1 H_p \cdot \partial_0 G_p = C(2d-3b)\,x^3 + C(4e-bd-6c)\,x^2y + C(6g+be-4cd)\,xy^2 + C(3bg-2ce)\,y^3.$$

Lean proof outline. Unfold definitions, apply `simp` for the multivariate derivative API, and finish by `ring`. Declaration: `binary_wronskian`. The coefficient formulas are exactly the scalar Stage 2 quantities w_3, w_2, w_1, w_0. □

4 Theorem–declaration mapping

#	Lean declaration	Mathematical content
1	`two_pair_decomposition`	Thm. 4: two-pair sextic identity
2	`split_pair_decomposition`	Thm. 5: $u^5v + v^6$ factorization
3	`outer_jacobian_det`	Thm. 6: outer 4×4 factor
4	`paired_jacobian_det`	Thm. 7: paired 4×4 factor
5	`section_root_ne_zero`	Thm. 8: $k \neq 0$
6	`leading_W_ne_zero`	Thm. 9: $2d - 3b \neq 0$

7	binary_wronskian	Thm. 10: formal Wronskian

Namespace: `Hodge.ExplicitResidueAlgebra`. Source file: `Hodge/ExplicitResidueAlgebra.lean`. No generated helper declarations from Stage 2 are split into this paper.

5 Axiom audit and verification evidence

A Stage 1 seven-declaration `\#print axioms` audit (`/workspace/hodge-formal-stage1-evidence-20261106/axioms.txt`, source `Stage1Audit.lean`) reports only the standard Lean axioms `propext`, `Classical.choice`, and `Quot.sound` (the first two theorems use only `propext` and `Quot.sound`). The combined Stage 1+Stage 2 audit (`ResidueAxiomAudit.lean`; log `residue-audit-e207394.log`) reports

`PASS: 52 incremental theorem declarations; standard axioms only.`

Repository checks record no `sorry`, `admit`, or `native_decide` in the incremental modules (`no-sorry-e207394.log`: "OK: no sorry/admit/native_decide, no axiom declarations").

Documented Stage 1 module build: `lakebuildHodge.ExplicitResidueAlgebra` passes with 7084 build targets (`docs/FULL_FORMALIZATION_STAGE1.md`). The Stage 2 incremental build that includes this module reports 7085/7085 success for `Hodge.ExplicitResidueCertificates` (`build-and-audit-final.log`). A full repository `lake build` at commit `e207394` completed successfully (`full-build-e207394.log`).

6 Toolchain pins and reproducibility

Pin	Value
Lean [2]	`leanprover/lean4:v4.22.0` (`lean-toolchain`)
mathlib input [3]	`v4.22.0` (`lakefile.toml`)
mathlib checkout	`79e94a093aff4a60fb1b1f92d9681e407124c2ca`
Git commit	`e207394a9ed4255ad804860b8f69fde7e940dd54`
PR	https://github.com/BenFrohman/HODGE/pull/36
License	Apache-2.0 (unchanged; no relicense)

Reproduction commands (from a checkout of the pinned commit):

```
lake exe cache get
lake build Hodge.ExplicitResidueAlgebra
python3 scripts/check_no_sorry.py
# Combined audit after Stage 2 is present:
lake build Hodge.ExplicitResidueAlgebra \
           Hodge.ExplicitResidueCertificates
python3 scripts/check_residue_axioms.py
```

Local evidence directories inspected while preparing this draft (read-only; another worker packages proof evidence):

- `/workspace/hodge-formal-stage1-20261106/` (mathematical source pin `e207394a9ed4255ad804860b8f69fde7e940dd54`; later inspected worktree `fefe2ba9d87c900b538d02c13f7f51f375fcd285` retains identical reviewed modules, audit, and toolchain/lock files);
- `/workspace/hodge-formal-stage1-evidence-20261106/` (`Stage1Audit.lean`, `axioms.txt`, `baseline-explicit-curve.lean`);
- `/workspace/hodge-formal-stage2-evidence-20261106/` and `/workspace/hodge-certificate-delivery-20261106/` (build/audit logs cited above).

CI verification at a later identical head. User-reported merge of PR https://github.com/BenFrohman/HODGE/pull/36 is acknowledged; this packaging records no merge-commit SHA (merge verification is underway separately). At head `fefe2ba9d87c900b538d02c13f7f51f375fcd285`, where the reviewed Lean modules, audit, and toolchain/lock match mathematical pin `e207394a9ed4255ad804860b8f69fde7e940dd54`, all five reported GitHub Actions checks completed successfully: Lean/build and Lean/attempt (https://github.com/BenFrohman/HODGE/actions/runs/42317080169), Lean certificate/certify on the PR (https://github.com/BenFrohman/HODGE/actions/runs/42317080616), Lean certificate/certify on the push (https://github.com/BenFrohman/HODGE/actions/runs/42317075046), and Paper/paper (https://github.com/BenFrohman/HODGE/actions/runs/42317080271). Both certification logs build the incremental modules and print `PASS: 52 incremental theorem declarations; standard axioms only`. CI success verifies those algebraic declarations; it does not discharge geometric Hodge/Betti/cycle-class obligations. This note remains a REVIEW DRAFT with Zenodo draft DOI `10.5281/zenodo.24025148` reserved; unpublished (not a live citation / not a publication claim).

7 Limitations and relation to the geometric program

The seven lemmas are genuine polynomial and matrix identities. They do not instantiate X as a smooth projective scheme, do not produce a Betti cycle class, and do not identify Jacobian characters with geometric Hodge classes. Villaflor Loyola's theorem [4] and Griffiths' Jacobian-ring comparison remain external classical input (`NOTICE.md`, `docs/FULL_FORMALIZATION_STAGE1.md`).

These algebraic notes can be fully formalized—and, for the seven lemmas above, already are—while publication of a full geometric Hodge paper for X remains held pending the missing geometric foundations. No novelty or peer-review claim is made.

Author, license, and contact

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References

[1] Benjamin Stanley Frohman. HODGE: Lean 4 formalization and hodge ledger. https://github.com/BenFrohman/HODGE, 2026. Source commit e207394a9ed4255ad804860b8f69fde7e940dd54; PR https://github.com/BenFrohman/HODGE/pull/36. Historical archive DOI 10.5281/zenodo.23629480 identifies prior repository provenance and is not a DOI for these incremental algebra notes.

[2] Lean FRO. The Lean 4 theorem prover. https://github.com/leanprover/lean4, 2026. Pinned release v4.22.0 via lean-toolchain.

[3] mathlib community. mathlib4. https://github.com/leanprover-community/mathlib4, 2026. Pinned input revision v4.22.0; locked checkout 79e94a093aff4a60fb1b1f92d9681e407124c2ca from lake-manifest.json.

[4] Roberto Villaflor Loyola. Periods of complete intersection algebraic cycles. *Manuscripta Mathematica*, 167:765–792, 2022. Cited as classical residue-cycle input; not formalized in the incremental Lean modules of this note.